

SENSORS

POSITION SENSITIVE DEVICE

- Sensors for *distance measurements* are among the most important ones in robotics
- They are used to measure distances to the nearest obstacle around the robot for navigation purposes
- In the past, most robots have been equipped with sonar sensors
- Because of the relatively *narrow cone* of these sensors, a typical configuration requires 24 sensors (mapping about 15° each)
- Sonar sensors use the following principle:
  - A short acoustic signal of about 1ms at an ultrasonic frequency of 50kHz to 250kHz is emitted and the time is measured from signal emission until the echo returns to the sensor.
  - The measured *time-of-flight* is proportional to twice the distance of the nearest obstacle in the sensor cone.
  - If no signal is received within a certain time limit, then no obstacle is detected within the corresponding distance.
  - Measurements are repeated about 20 times per second, which gives this sensor its typical *clicking sound*

SONAR SENSOR

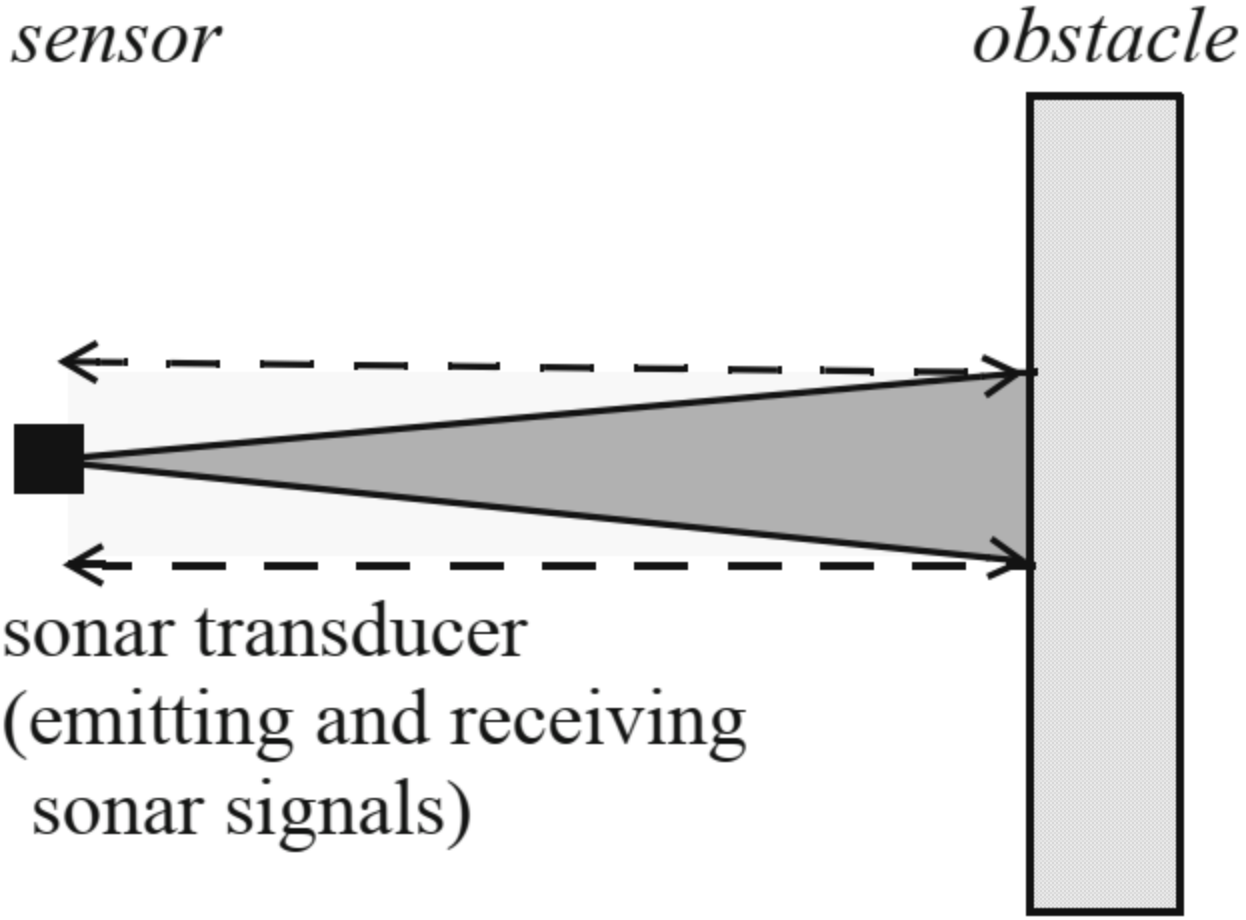


Figure 3.5: Sonar sensor

SONAR SENSOR



SONAR SENSOR



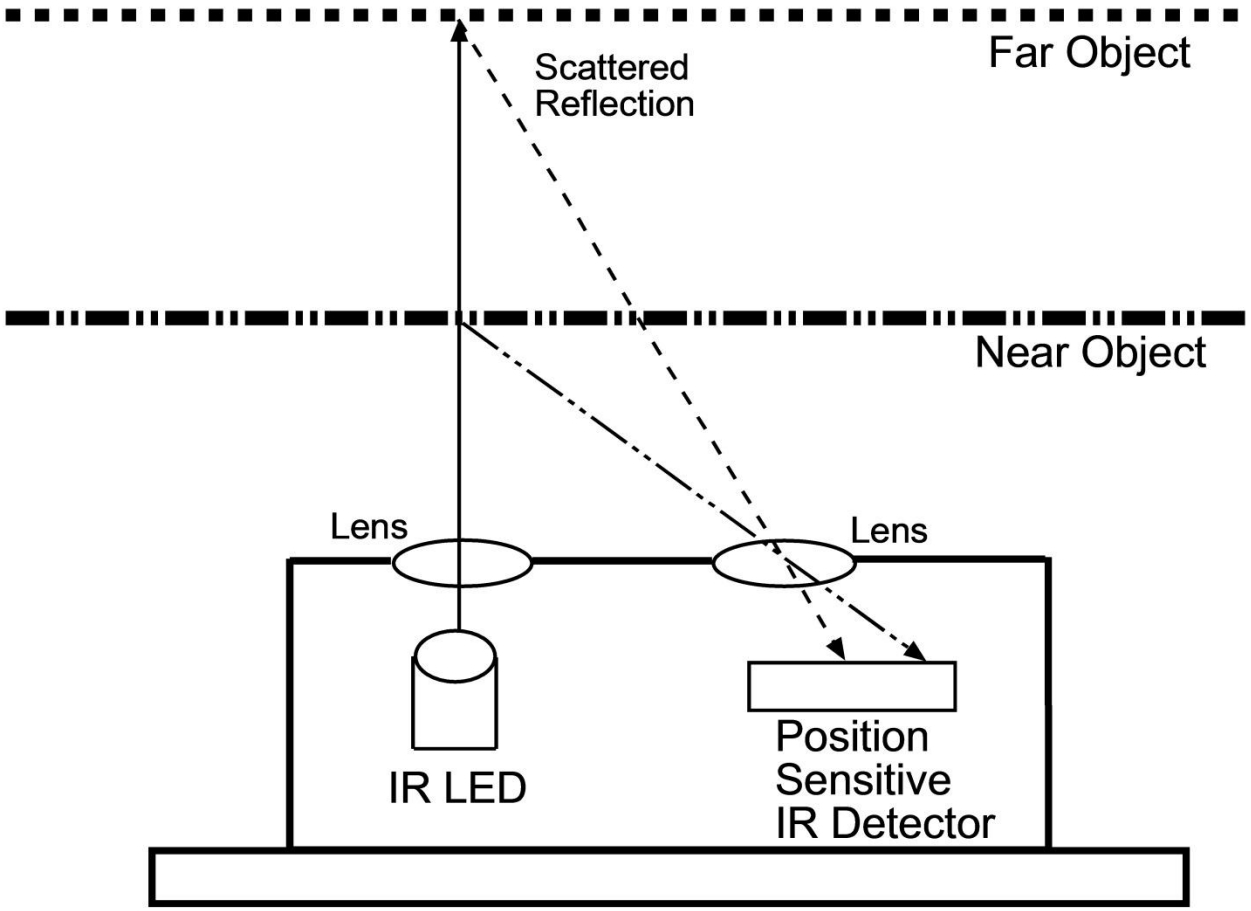
SONAR SENSOR

- Sonar sensors have a number of disadvantages but are also a very powerful sensor system
- The most significant problems of sonar sensors are narrow range, reflections and interference.
- When the acoustic signal is reflected, for example off a wall at a certain angle, then an obstacle seems to be further away than the actual wall that reflected the signal.
- Interference occurs when several sonar sensors are operated at once (among the 24 sensors of one robot, or among several independent robots).
- Here, it can happen that the acoustic signal from one sensor is being picked up by another sensor
- Coded sonar signals can be used to prevent this, for example using pseudo random codes [Jörg, Berg 1998].

INFRARED SENSOR

- Today, in many mobile robot systems, sonar sensors have been replaced by either infrared sensors or laser sensors
- The current standard for mobile robots is laser sensors that return an almost perfect local 2D map from the viewpoint of the robot, or even a complete 3D distance map
- Unfortunately, these sensors are still too large and heavy (and too expensive) for small mobile robot systems.
- Infrared (IR) distance sensors do not follow the same principle as sonar sensors, since the time-of-flight for a photon would be much too short to measure with a simple and cheap sensor arrangement.
- Instead, these systems typically use a pulsed infrared LED at about 40kHz together with a detection array.
- The angle under which the reflected beam is received changes according to the distance to the object and therefore can be used as a measure of the distance.
- The wavelength used is typically 880nm.
- Although this is invisible to the human eye, it can be transformed to visible light either by IR detector cards or by recording the light beam with an IR-sensitive camera.

SENSOR CATEGORIES



SENSOR CATEGORIES

NI myRIO Project Essentials Guide

IR Range Finder

- SHARP GP2Y0A21YK0F
- features and connections
- principle of operation
- calibration techniques



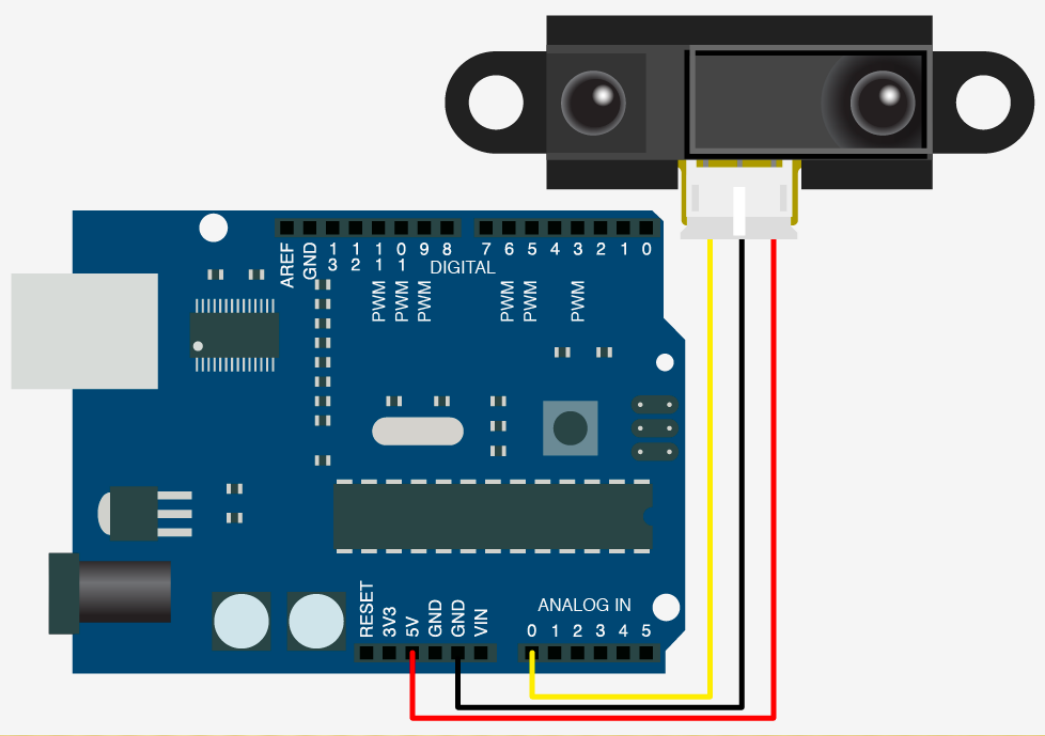
POWERED BY

LabVIEW

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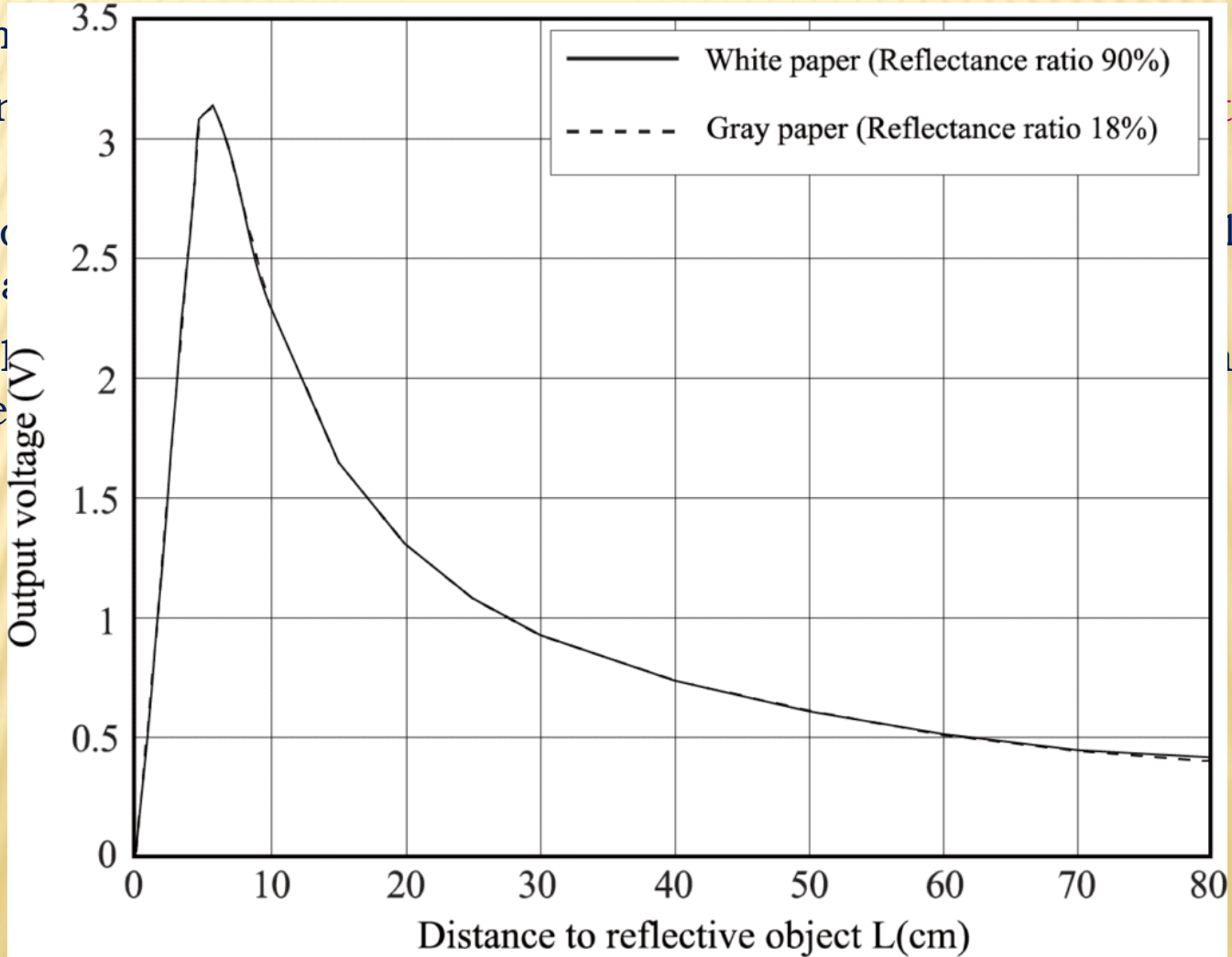
IR SENSOR INTERFACE TO ARDUINO

- There are two variations of this sensor:
  - ✓ Sharp GP2D12 with analog output
  - ✓ Sharp GP2D02 with digital serial output
- The analog sensor simply returns a voltage level in relation to the measured distance (unfortunately not proportional).
- The digital sensor has a digital serial interface.
- It transmits an 8bit measurement value bit-wise over a single line, triggered by a clock signal from the CPU.



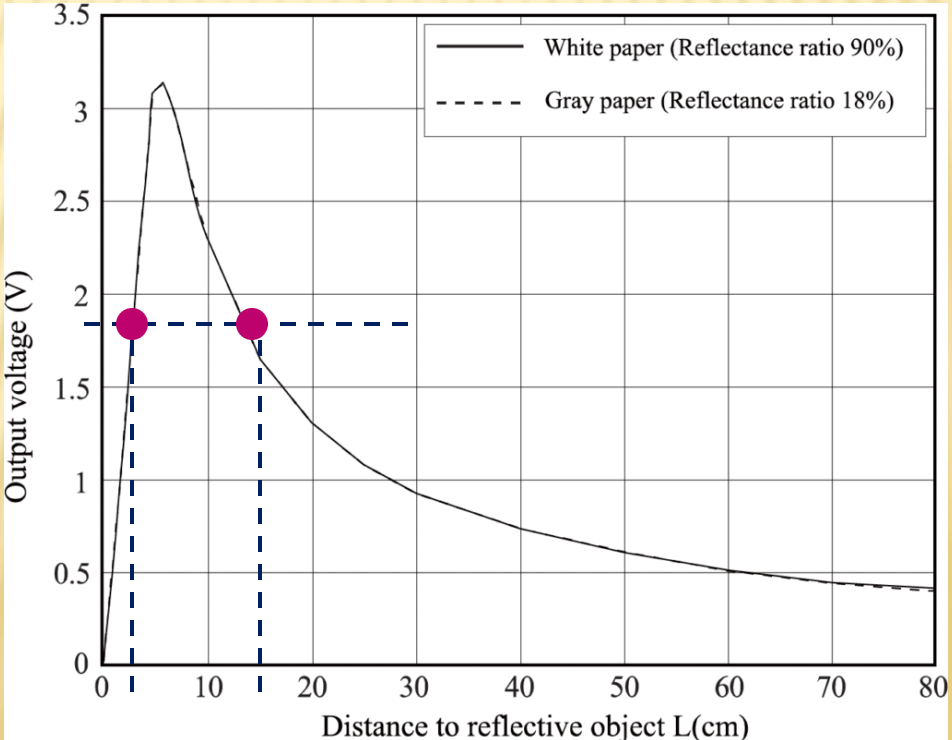
INFRARED SENSOR

- The relationship between digital sensor read-out (raw data) and actual distance information as can be seen *not linear*
- So som
- The sir ...able which can be
- Since ... ll have the reasona
- With t... sor and is comple



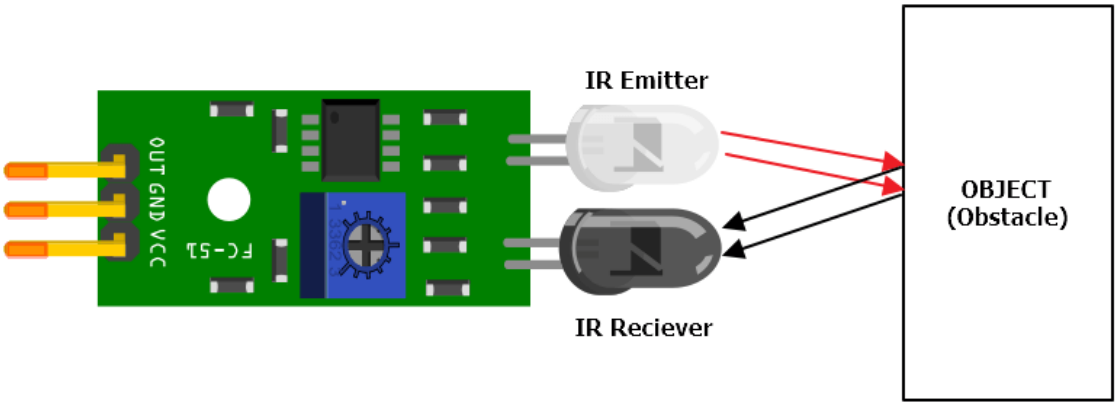
INFRARED SENSOR

- Another problem becomes evident when looking at the diagram for actual distances *below about 6cm*.
- These distances are below the measurement range of this sensor and will result in an incorrect reading of a higher distance.
- This is a more serious problem, since it cannot be fixed in a simple way.
- The safest solution is to mechanically mount the sensor in such a way that an obstacle can never get closer than 6cm.



IR PROXIMITY SENSOR

- IR proximity switches are of a much simpler nature than IR PSDs.
- IR proximity switches are an electronic equivalent of the tactile binary sensors.
- These sensors also return only 0 or 1, depending on whether there is free space (for example 1-2cm) in front of the sensor or not.
- IR proximity switches can be used instead of tactile sensors for most applications that involve obstacles with reflective surfaces.
- They also have the advantage that no moving parts are involved compared to mechanical micro-switches.



Note: Black surfaces absorbs light naturally. So it will not reflects much light on IR Reciever. This concept is used in Line follower robot.

COMPASS SENSOR

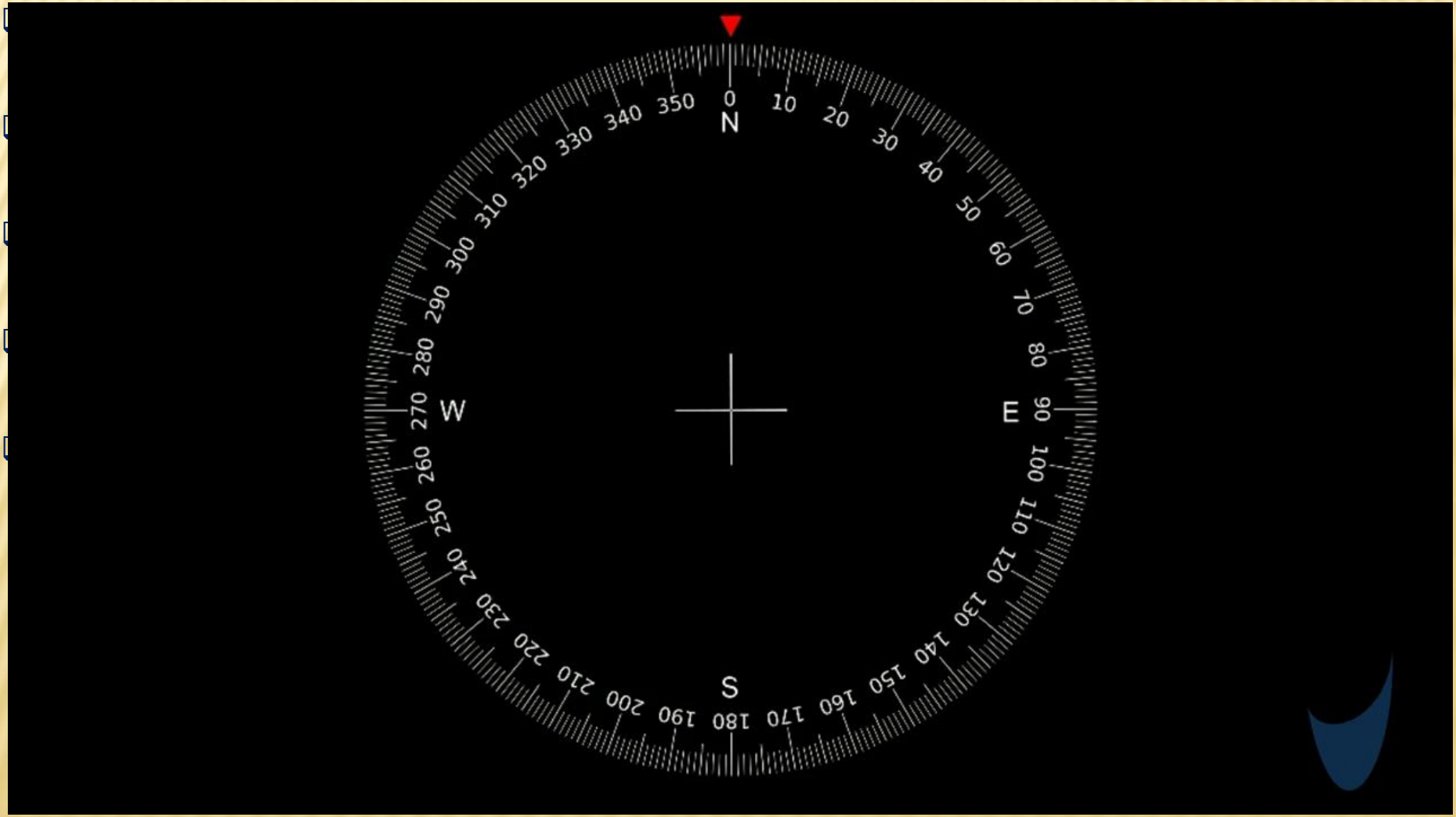
- A compass is a very useful sensor in many mobile robot applications, especially self-localization.
- An autonomous robot has to rely on its on-board sensors in order to keep track of its current position and orientation.
- The standard method for achieving this in a driving robot is to use shaft encoders on each wheel, then apply a method called "*dead reckoning*".
- This method starts with a known initial position and orientation, then adds all driving and turning actions to find the robot's current position and orientation.
- Unfortunately, due to wheel slippage and other factors, the "dead reckoning" error will grow larger and larger over time
- Therefore, it is a good idea to have a compass sensor onboard, to be able to determine the robot's absolute orientation
- A further step in the direction of global sensors would be the interfacing to a receiver module for the satellite-based global positioning system (GPS).
- GPS modules only work outdoors in unobstructed areas

COMPASS SENSOR

- Several compass modules are available for integration with a controller.
- The simplest modules are analog compasses that can only distinguish eight directions, which are represented by different voltage levels.
- These are rather cheap sensors, which are, for example, used as directional compass indicators in some four-wheel-drive car models.
- Such a compass can simply be connected to an analog input of the EyeBot and thresholds can be set to distinguish the eight directions.
- Digital compasses are considerably more complex, but also provide a much higher directional resolution.
- The sensor we selected for most of our projects has a resolution of 1° and accuracy of 2°, and it can be used indoors

COMPASS SENSOR

- Several compass modules are available for integration with a controller.



GYROSCOPE, ACCELEROMETER, INCLINOMETER

- Orientation sensors determine robot's orientation in 3D space
- They are required for tracked robots (Chapter 8), balancing robots (Chapter 10), walking robots (Chapter 11), or autonomous planes (Chapter 12).
- We will concentrate on sensors capable of measuring a single dimension.
- Two or three sensors of the same model can be combined for measuring two or all three axes of orientation.
- Sensor
  - Acc
    - ✓ A
    - ✓ A
  - Gyr ut one axis
    - ✓ H
  - Incli one axis
    - ✓ Seika N3 (analog output)
    - ✓ Seika N3d (PWM output)





ACCELEROMETER & GYROSCOPE

- Accelerometer measures accelerations "a" in x-y-z
- Gyroscope measures angular "ω" velocity around x-y-z
- Both are called inertial sensors/trackers
- Modern inertial trackers are solid-state structures that use micro-electro-mechanical systems(MEMS) technology.
- Three gyroscopes are machined on mutually orthogonal axes, measuring *yaw*, *pitch*, and *roll* angular velocities ω.
- The orientation angle θ about the three orthogonal axes is then determined through integration over time.
- Digitally; integration is done using accumulator "Summing"

$$\theta_x = \int \omega_x dt$$

$$\theta_y = \int \omega_y dt$$

$$\theta_z = \int \omega_z dt$$



$$\theta_x(nT) = \sum_{k=0}^{n} \omega_x(k)T$$

$$\theta_x(nT) = T\sum_{k=0}^{n} \omega_x(k)$$

ACCELEROMETER & GYROSCOPE

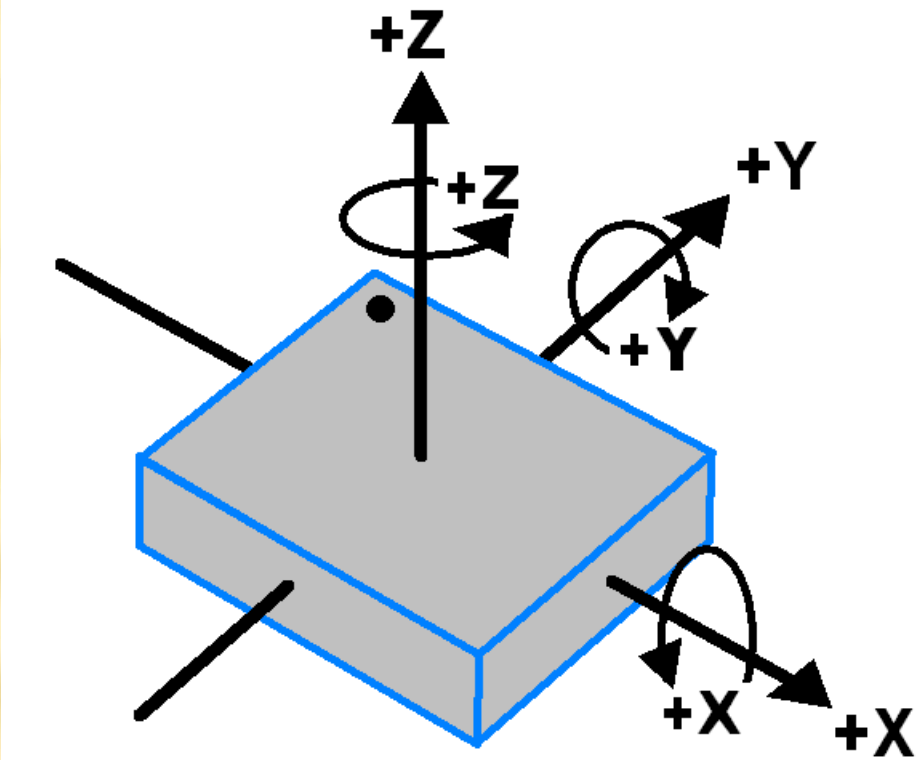
- Three accelerometers machined with the three gyroscopes are needed to measure body referenced accelerations.
- Knowing the tracked object orientation (from gyroscopic data) and subtracting gravitational acceleration allows the computation of accelerations in world coordinates.
- The tracked object position is finally obtained by double integration over time and knowledge of starting position (calibration).

$$x = \iint a_x dt$$

$$y = \iint a_y dt$$

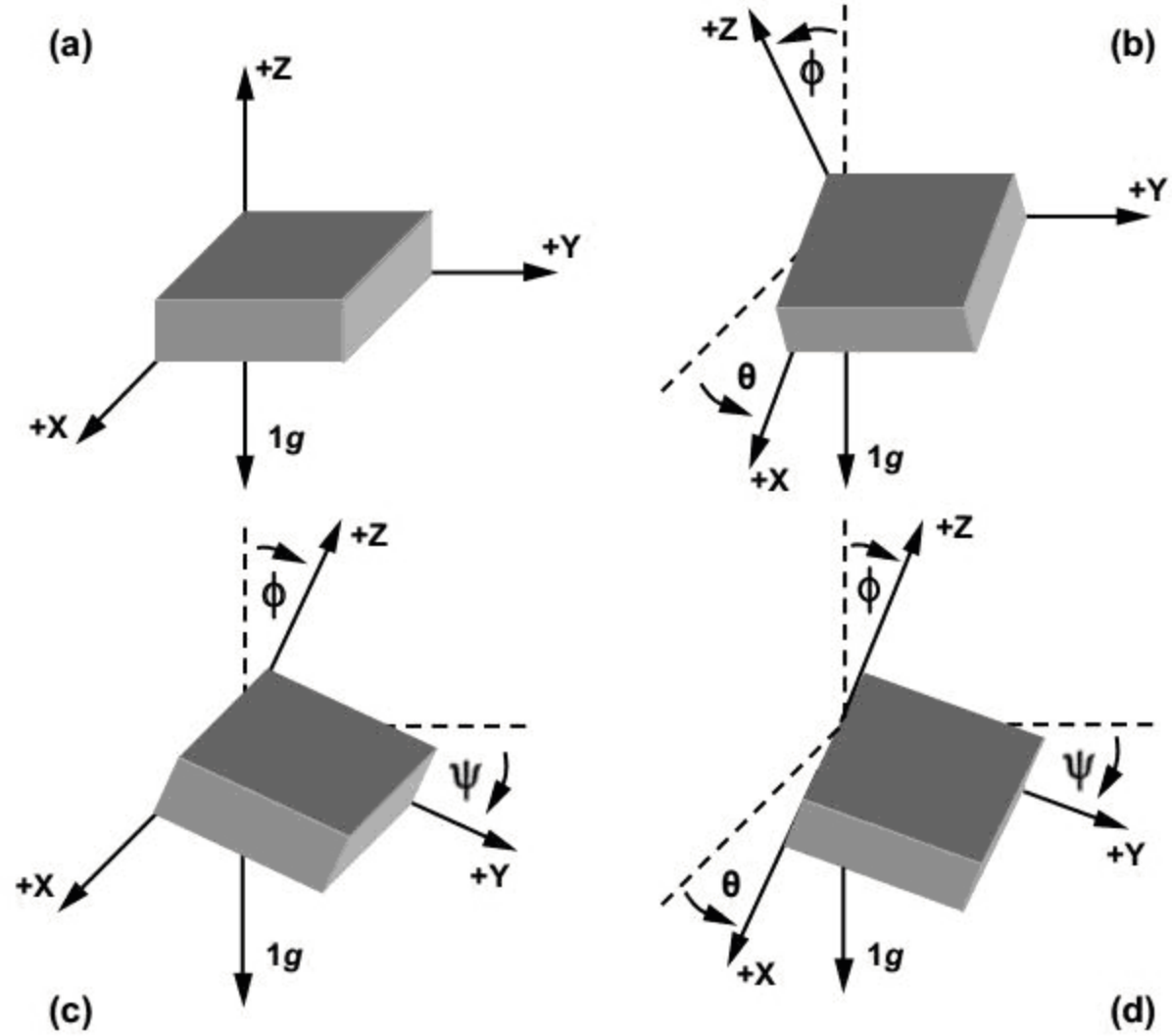
$$z = \iint a_z dt$$

ACCELEROMETER & GYROSCOPE

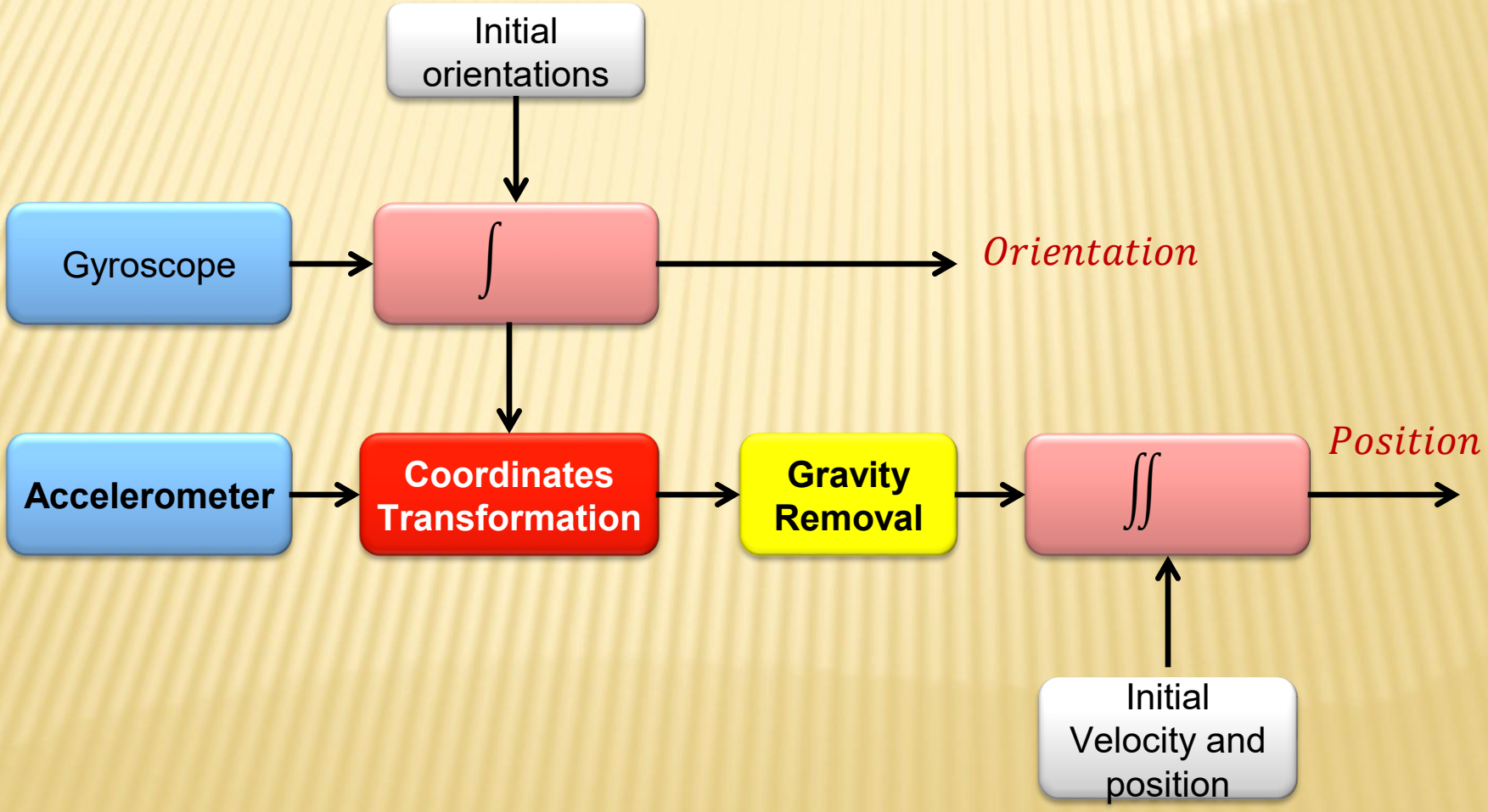


MPU-6050
Orientation & Polarity of Rotation

ACCELEROMETER COORDINATE TRANSFORMATION

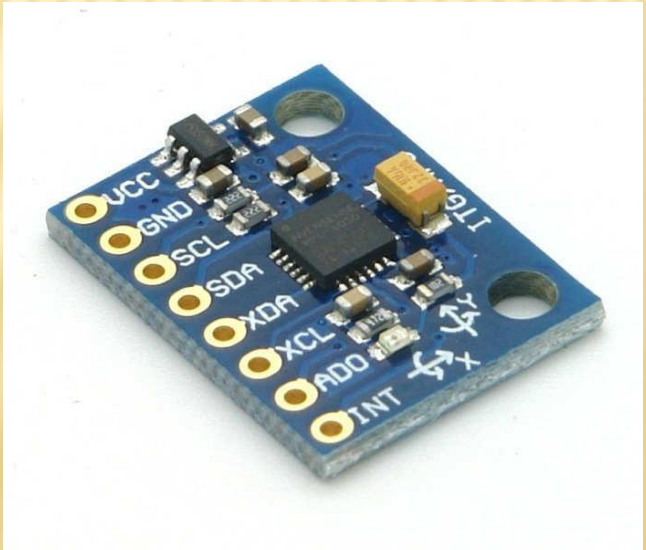


ACCELEROMETER COORDINATE TRANSFORMATION



MPU6050

- MPU6050 sensor module is complete 6-axis Motion Tracking Device.
- It combines 3-axis Gyroscope, 3-axis Accelerometer and Digital Motion Processor all in small package.
- Also, it has additional feature of on-chip Temperature sensor.
- It has I2C bus interface to communicate with the microcontrollers.
- It has Auxiliary I2C bus to communicate with other sensor devices like 3-axis Magnetometer, Pressure sensor etc.
- If 3-axis Magnetometer is connected to auxiliary I2C bus, then MPU6050 can provide complete 9-axis Motion Fusion output.



INCLINOMETER

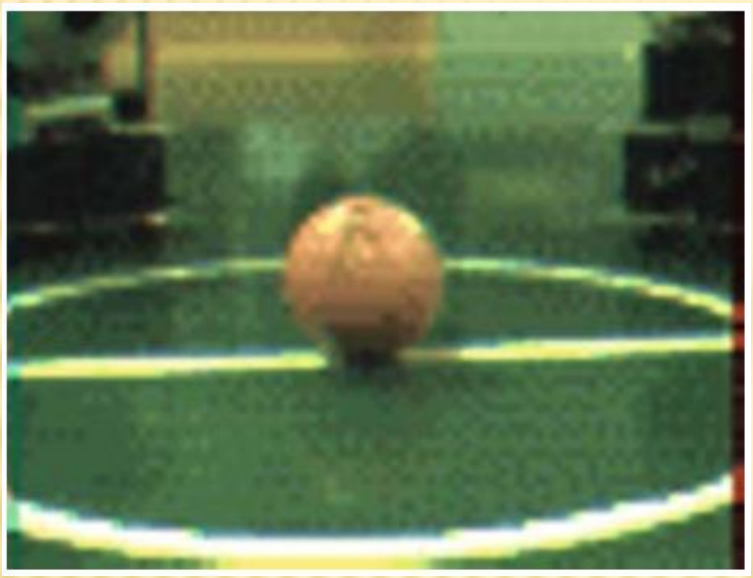
- Inclinometers (tilt sensor) measure the absolute orientation angle within a specified range, depending on the sensor model.
- The sensor output is either analog signal output or PWM.
- Since inclinometers measure the absolute orientation angle about an axis and not the derivative, they seem to be much better suited for orientation measurement than a gyroscope.
- However, our measurements with the Seika inclinometer showed that they suffer a time lag when measuring and also are prone to oscillation when subjected to positional noise, for example as caused by servo jitter.
- Especially in systems that require immediate response, for example balancing robots in Chapter 10, *gyroscopes have an advantage over inclinometers*.
- The ideal solution was a combination of inclinometer and gyroscope

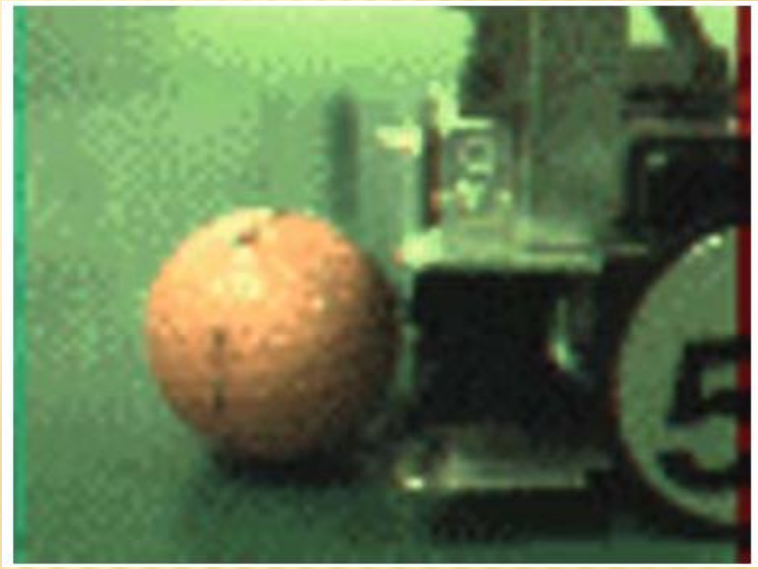
CAMERA SENSOR

DIGITAL CAMERA

- Digital cameras are the most complex sensors used in robotics.
- They have not been used in embedded systems until recently, because of the processor speed and memory capacity required
- For mobile robot applications, we are interested in a high frame rate, because our robot is moving and we want updated sensor data as fast as possible.
- Since there is always a trade-off between high frame rate and high resolution, we are not so much concerned with camera resolution.
- For most applications for small mobile robots, a resolution of 60× 80 pixels is sufficient.
- Even from such a small resolution we can detect, for example, colored objects or obstacles in the way of a robot (see 60× 80 sample images from robot soccer)

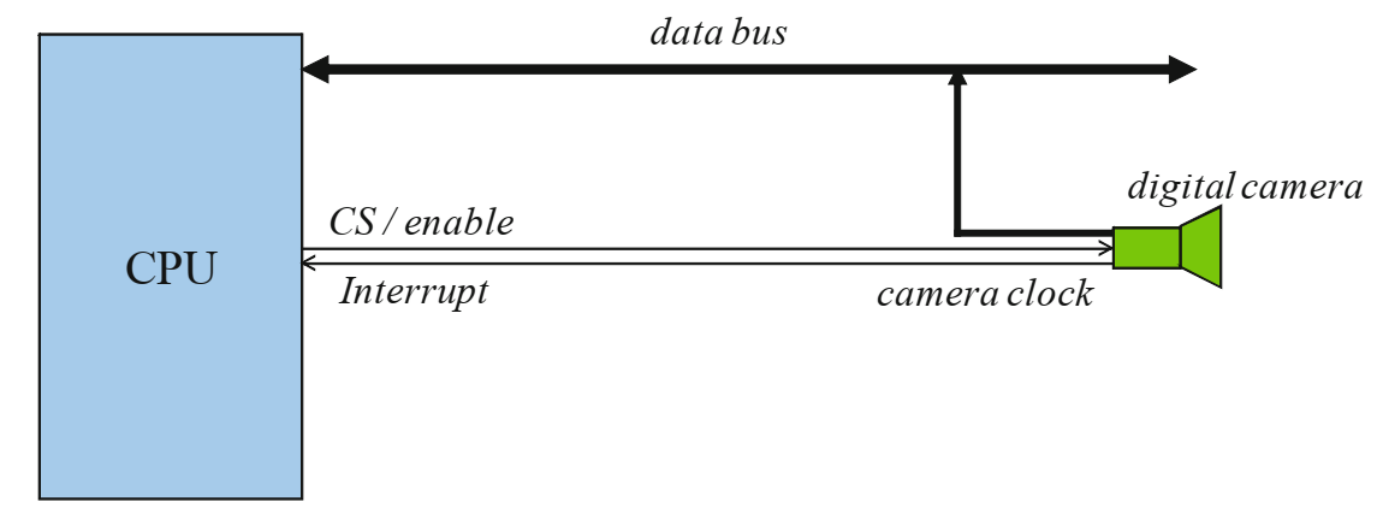
DIGITAL CAMERA





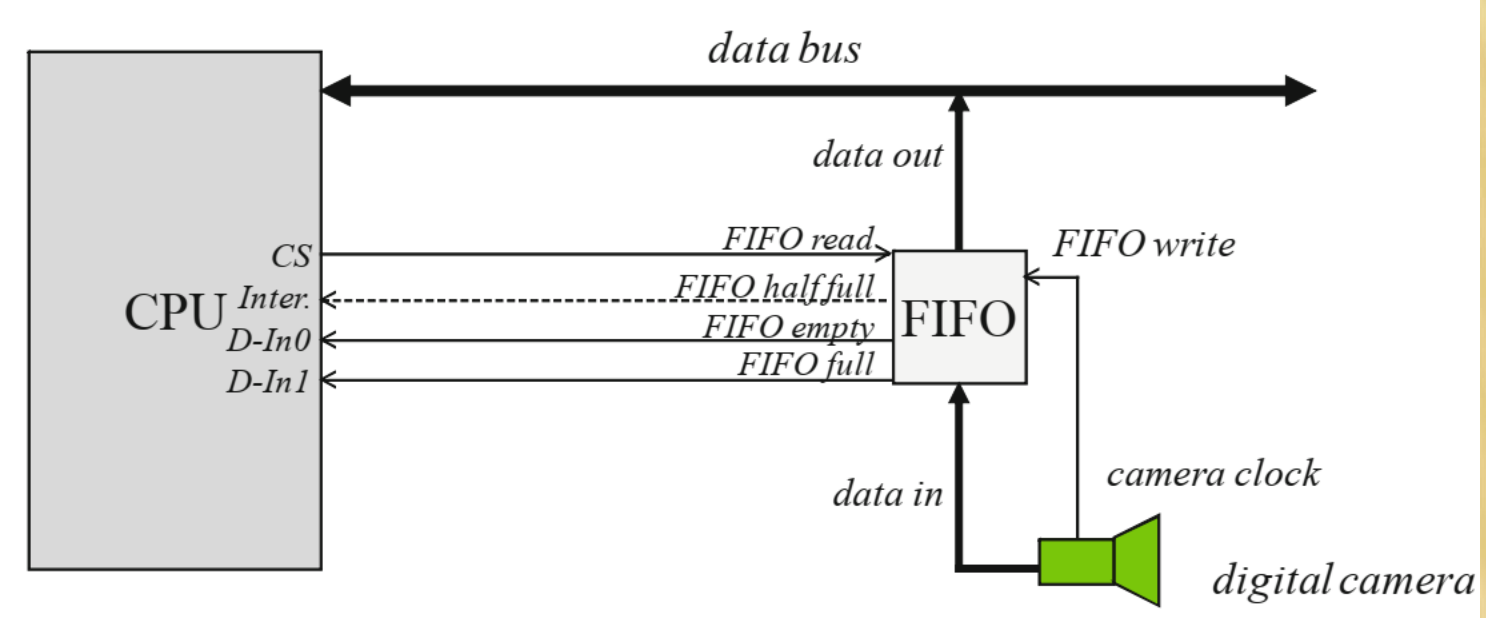
CAMERA INTERFACE

- The simplest camera interface to a CPU is shown.
- The camera clock is linked to a CPU interrupt, while the parallel camera data output is connected directly to the data bus.
- Every single image byte from the camera will cause an interrupt at the CPU, which will then enable the camera output and read one image data byte from the data bus
- Every interrupt creates considerable overhead, since system registers have to be saved and restored on the stack.
- Therefore, creating one interrupt per image byte is not the best possible solution



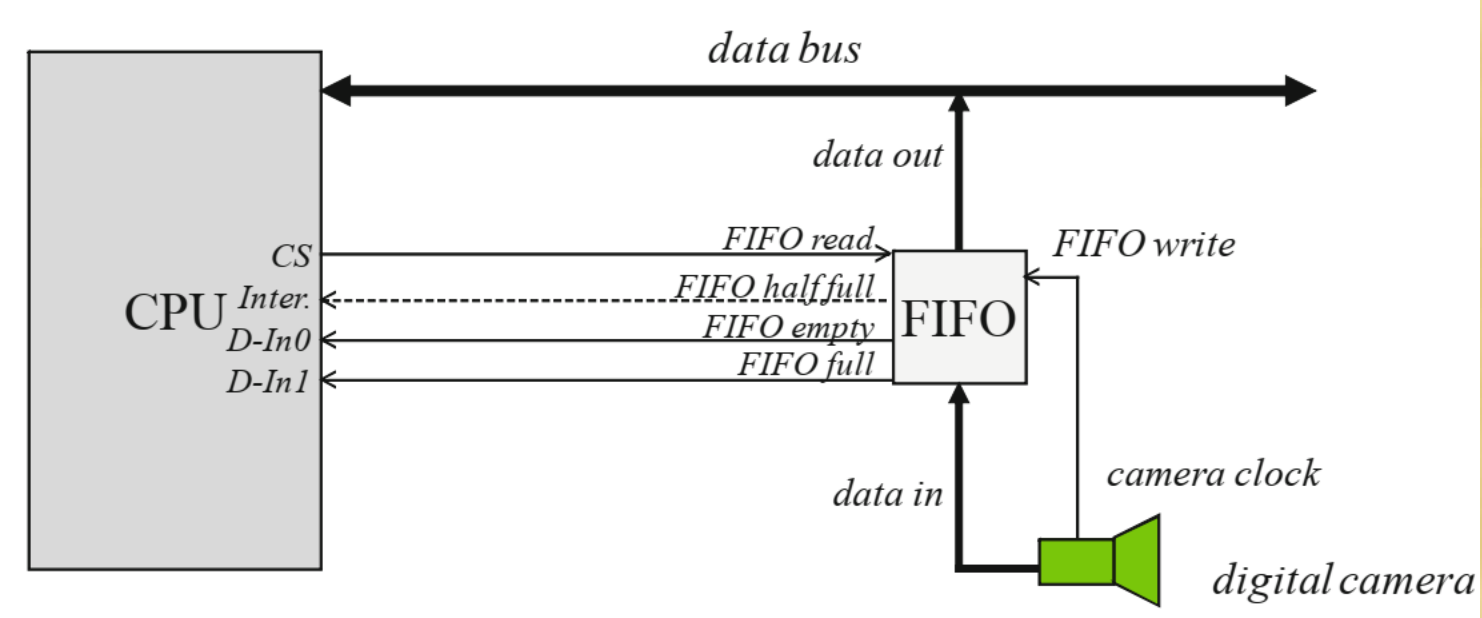
CAMERA INTERFACE

- It would be better to buffer a number of bytes and then use an interrupt much less frequently to do a bulk data transfer of image.
- Figure shows this approach using a FIFO buffer for intermediate storing of image data.
- The advantage of a FIFO buffer is that it supports unsynchronized read and write in parallel.
- So while the camera is writing data to the FIFO buffer, the CPU can read data out, with the remaining buffer contents staying undisturbed.



CAMERA INTERFACE

- The camera output is linked to the FIFO input, with the camera's pixel clock triggering the FIFO write line.
- From the CPU side, the FIFO data output is connected to the system's data bus, with the chip select triggering the FIFO read line.
- The FIFO provides three additional status lines
  - ✓ Empty flag
  - ✓ Full flag
  - ✓ Half full flag



CAMERA INTERFACE

- These digital outputs of the FIFO can be used to control the bulk reading of data from the FIFO.
- Since there is a continuous data stream going into the FIFO, the most important of these lines in our application is the half full flag, which we connected to a CPU interrupt line.
- Whenever the FIFO is half full, we initiate a bulk read operation of 50% of the FIFO's contents.
- Assuming the CPU responds quickly enough, the full flag should never be activated, since this would indicate an imminent loss of image data

CAMERA DRIVER

- There are three commonly used capture modes available for receiving data from a digital camera:
- Read mode: The application requests a frame from the driver and blocks CPU execution.
- The driver waits for the next complete frame from the camera and captures it.
- Once a frame has been completely read in, the data is passed to the application and the application continues.
- In this mode, the driver will first have to wait for the new frame to start.
- This means that the application will be blocked for up to two frames, one to find the start of a new frame and one to read the current frame.

CAMERA DRIVER

- Continuous capture mode: In this mode, the driver continuously captures a frame from the camera and stores it in one of two buffers.
- A pointer to the last buffer read in is passed to the application when the application requests a frame.
- Synchronous continuous capture mode: In this mode, the driver is working in the background.
- It receives every frame from the camera and stores it in a buffer.
- When a frame has been completely read in, a trap signal/software interrupt is sent to the application.
- The application's signal handler then processes the data.
- The processing time of the interrupt handler is limited by the acquisition time for one camera image.